

Flat panel type multianode PMT assembly H13700 series



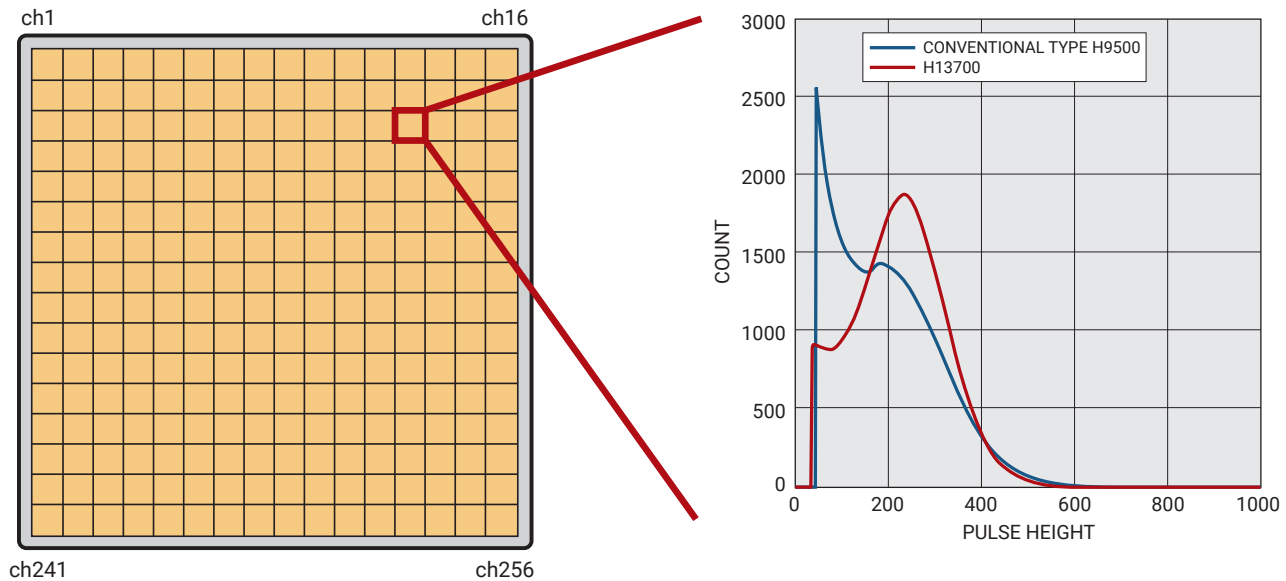
Features

- Wide effective area: 48.5 mm × 48.5 mm
- Packing density: 88 %
- 16 × 16 multianode, pixel size: 3 mm × 3 mm / anode
- High quantum efficiency: 33 % Typ.
- Small dead space
- Fast time response
- Cable HV input type

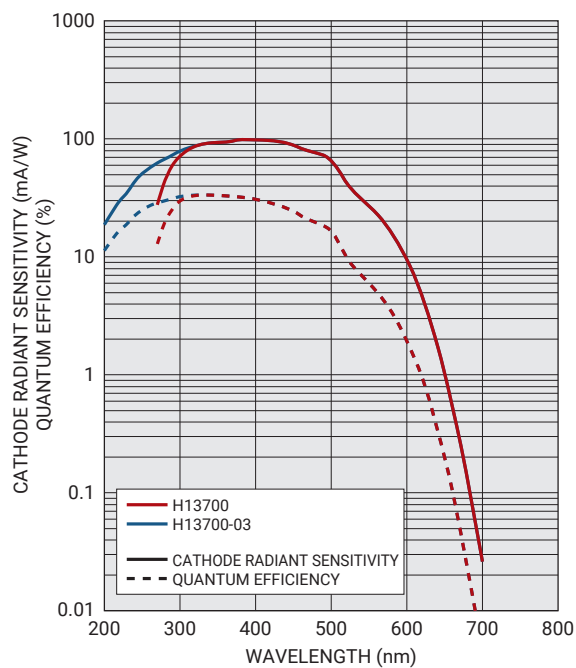
Applications

- Academic research (RICH, gamma ray telescope, etc.)
- Nuclear medicine equipment (PET, gamma camera, etc.)
- Neutron imaging

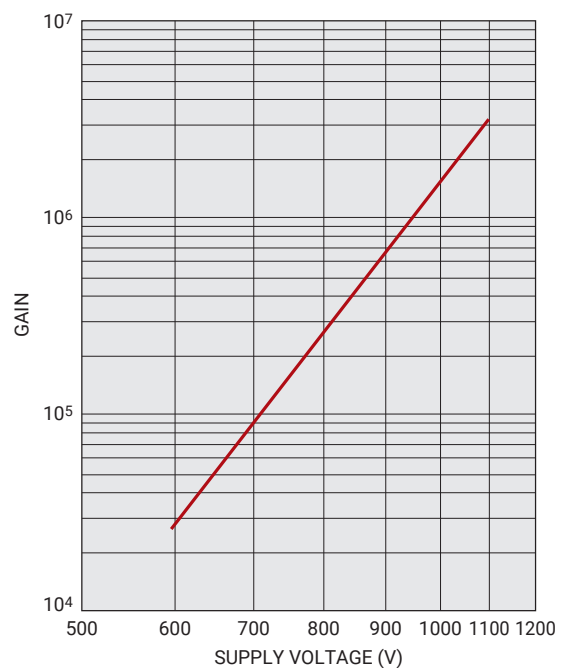
Single photon counting (example)



Typical spectral response



Typical gain



Specifications

Parameter			H13700	H13700-03	Unit	
Spectral response range			300 to 650	185 to 650	nm	
Peak wavelength			400			
Photocathode material			Bialkali		—	
Window material			Borosilicate	UV	—	
Dynode	Structure		Metal channel		—	
	Stage		10		—	
Maximum ratings	Supply voltage between anode and cathode		-1100		V	
	Average anode output current in total		100		μA	
	Divider current at -1100 V		185			
Cathode	Luminous sensitivity ①	Min.	60		μA/lm	
		Typ.	75			
		Blue sensitivity index ②	Typ.	12		—
Anode to cathode supply voltage			-1000		V	
Anode	Luminous sensitivity ③		Typ.	110	A/lm	
	Gain ③		Typ.	1.5 × 10 ⁶	—	
	Dark current per channel ④		Typ.	0.02	nA	
	Dark current in total ④	Typ.	5		nA	
		Max.	50			
	Time response ⑤	Rise time ⑥	Typ.	0.45		ns
		Transit time ⑦	Typ.	5.2		
		Transit time spread ⑧	Typ.	0.38		
Pulse linearity per channel at ±2 % deviation			Typ.	0.15	mA	
Uniformity between each anode			Typ.	1 : 3	—	
			Max.	1 : 5	—	

①The light source is a tungsten filament lamp operated at a distribution temperature of 2856 K. Supply voltage is 150 V between the cathode and all other electrodes connected together as anode.

②The value is cathode output current when a blue filter is interposed between the light source and the tube under the same condition as Note ①.

③Measured with the same light source as Note ① and with the anode-to-cathode supply voltage and voltage distribution ratio shown in "Voltage distribution ratio and supply voltage" below.

④Measured with the same supply voltage and voltage distribution ratio as Note ③ after 30 min storage in darkness.

⑤Those are test data when a signal from a central channel (P120) of 256 anodes is used, while all photocathode are illuminated by pulsed light source.

⑥The rise time is the time for the output pulse to rise from 10 % to 90 % of the peak amplitude when the whole photocathode is illuminated by a delta function light pulse.

⑦The electron transit time is the interval between the arrival of delta function light pulse at the entrance window of the tube and the time when the anode output reaches the peak amplitude. In measurement, the whole photocathode is illuminated.

⑧Also called transit time jitter. This is the fluctuation in electron transit time between individual pulses in the single photoelectron event, and defined as the FWHM of the frequency distribution of electron transit time.

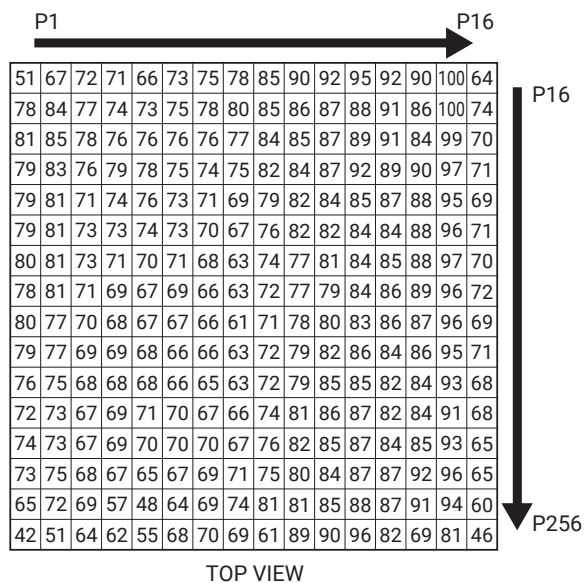
Voltage distribution ratio and supply voltage

Electrodes	K	Dy1	Dy2	Dy3	Dy4	Dy5	Dy6	Dy7	Dy8	Dy9	Dy10	GR	P
Distribution ratio	2	1	1	1	1	1	1	1	1	1	1	0.5	

Supply voltage: -1000 V, K: Cathode, Dy: Dynode, GR: Guard ring P: Anode

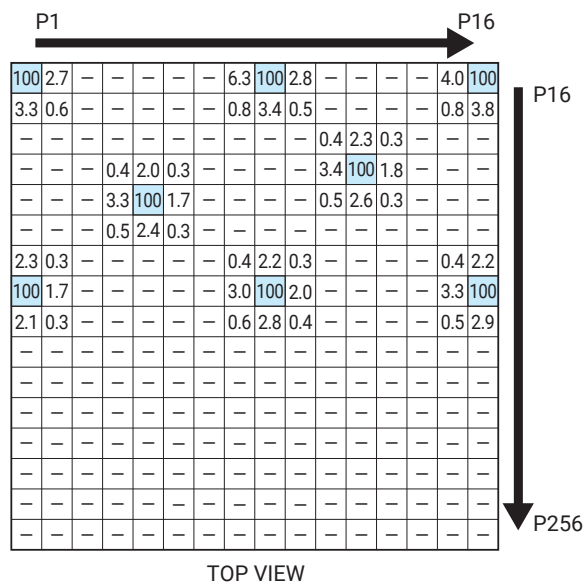
Characteristics

Anode uniformity (Example)



Supply voltage: -1000 V
Light source: Tungsten lamp with blue filter (DC Light)
Spot illumination (Aperture size): 3 mm square on each channel

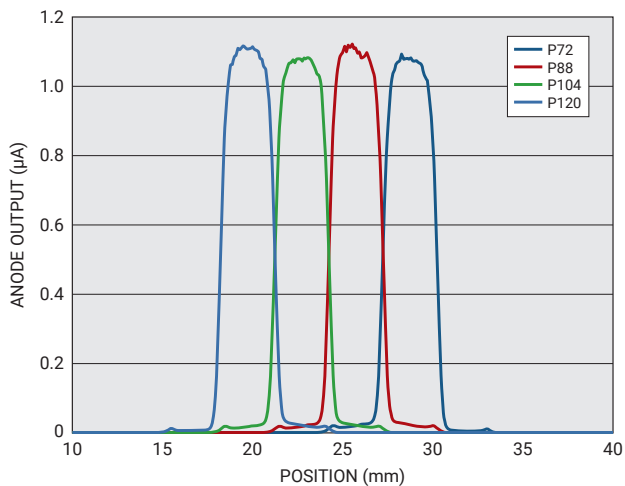
Anode cross-talk (Example)



Supply voltage: -1000 V
Light source: Tungsten lamp with blue filter (DC Light)
Fiber size: $\Phi 1.0$ mm (Kuraray: clear fiber NA=0.72)

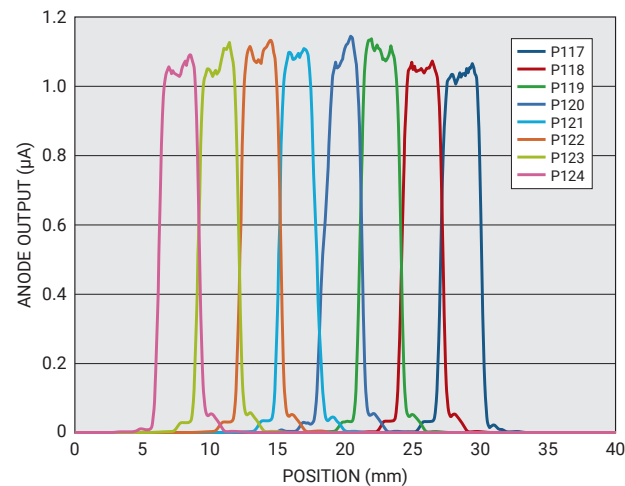
Cross uniformity (Example)

X-axis



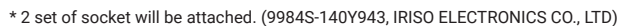
Supply voltage: -1000 V
Spot size: $\Phi 1.0$ mm

Y-axis



Supply voltage: -1000 V
Spot size: $\Phi 1.0$ mm

Dimensional outline (Unit:mm)



Schematic diagram of the HV divider circuit. The circuit includes a 100 kV SHV-P source connected to a series of resistors (R19, R1, R2-R7, R14-R17, R18) and capacitors (C1-C4). A feedback network with R20, R21, C11, and C12 is connected to the divider output. The divider output is connected to a 2-(DOUBLE-ROW 0.5 PITCH) CONNECTOR. The connector pins are labeled: P16, P15, P2, P1, P10 OUTPUT, ANODE OUTPUT (P1), ANODE OUTPUT (P2), ANODE OUTPUT (P15), ANODE OUTPUT (P16), ANODE OUTPUT (P241), ANODE OUTPUT (P242), ANODE OUTPUT (P255), ANODE OUTPUT (P256). The diagram also shows a ground connection (GR) and a signal ground (SIGNAL GND).

Component values:

- R1, R19: 1.0 M Ω
- R2-R7: 470 k Ω
- R14, R18: 10 k Ω
- R15 to R17: 51 Ω
- R20, R21: 4.99 k Ω
- C1, C2: 0.010 μ F
- C3: 0.022 μ F
- C4: 0.033 μ F
- C10: 0.010 μ F
- C11, C12: 0.0015 μ F

DIVIDER CURRENT $\approx$ 185 μ A at -1100 V (MAX.)

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